CAUÃ FELYPE CAETANO REIS

Junior Software Engineer · AI Systems & Code Evaluation · Polyglot Developer

Itaguaru, GO — Brazil (Remote / Worldwide) +55 (62) 9 9452-1953 caua.reis.dev@gmail.com

linkedin.com/in/cauafelypereis github.com/cauacreis English: Advanced

PROFESSIONAL SUMMARY

Junior Software Engineer and researcher with a proven background in **system architecture, algorithmic correctness, and AI model evaluation**. Co-author of an advanced scientific research paper submitted to the **IEEE Transactions on Software Engineering (IEEE TSE)** focused on Explainable AI (XAI/SHAP), Multi-Agent Systems (MAS), and Human-in-the-Loop (HITL) active learning. Polyglot practitioner fluent across **Python, Java 21, C, TypeScript, and SQL**, ranging from bare-metal x86 microkernel development to 24/7 high-concurrency production microservices. Experienced in designing rigorous test suites, edge-case analysis, reference solutions, and evaluating code quality, architectural trade-offs, and LLM reasoning. Advanced technical English proficiency.

CORE COMPETENCIES & CODE EVALUATION SKILLS

Code Review & Quality:	Syntax verification, time/space complexity analysis (Big-O), design patterns, bug triage, edge-case discovery, defensive programming, static analysis
AI & Model Evaluation:	Explainable AI (XAI via SHAP), Human-in-the-Loop (HITL) workflows, benchmark creation, LLM reasoning audit, prompt evaluation, multi-agent orchestration
Languages:	Python 3.11+, Java 21/17, C, TypeScript/JavaScript, SQL (PostgreSQL, MySQL), Assembly (x86 GAS)
System Architecture:	Distributed Microservices, AsyncIO Event Loops, Concurrency & Virtual Threads, WebSocket Streaming, RESTful APIs, OOP, Clean Code
Testing & Tooling:	Unit Testing, PyTest, JUnit, Test Case Synthesis, Git/GitHub, Docker, Linux, CI/CD Actions

RELEVANT EXPERIENCE & RESEARCH

OrenAI — GeoAI Platform & Scientific Research (IEEE TSE Paper) XAI / SHAP

github.com/cauacreis/OrenAI

Technical Lead & AI Architecture Researcher

- **Model Evaluation & Explainable AI (XAI):** Co-developed an ML framework combining Gradient Boosting (XGBoost/LightGBM) with CNNs, implementing **SHAP (SHapley Additive exPlanations)** to mathematically deconstruct 100% of model feature attributions, eliminating black-box outputs.
- **Human-in-the-Loop (HITL) Active Learning:** Architected feedback loops allowing human domain experts to evaluate, validate, or reject model inference, routing flagged samples to an active learning pipeline for incremental retraining.
- **IEEE TSE Publication:** Co-authored a full-length technical paper submitted to *IEEE Transactions on Software Engineering*, authoring architectural specifications, mathematical formalizations, and benchmarking methodology.

SKZ Trading — Production Algorithmic Trading Platform (24/7) Production 24/7

Jul 2026 – Present | skztrading.com

Software Architect & Backend Developer

- **Asynchronous Execution & Reliability:** Engineered 11 asynchronous Python microservices (`asyncio`, `aiohttp`) operating continuously on Linux VPS with 99.9% uptime, processing real-time market data under sub-100ms latency.
- **Defensive Logic & Edge Cases:** Developed a proprietary multi-layer liquidity scoring algorithm (0–100) with strict margin verification floors, ensuring zero transactions execute below cost thresholds under volatile market conditions.
- **Security Hardening:** Conducted rigorous OWASP Top 10 code audits, migrating asymmetric key decryption (Ed25519/AES) locally to eliminate plain-text key exposure in API payloads.

MicroOS — x86 32-bit Microkernel Written from Scratch in C C / Systems

github.com/cauacreis/MicroOS

Systems Programmer

- Designed and verified a 32-bit operating system kernel from scratch (C & Assembly), implementing two-level paging, Ring 3 isolation, preemptive round-robin scheduling (PIT 100Hz), and IPC message queues.
- Demonstrated mastery over fundamental low-level computing: memory allocation, context switching, interrupt handlers (IDT), and hardware-level debugging under QEMU.

CoreSync CRM & NexoTrade-OS — High-Concurrency Java 21 Architectures github.com/cauacreis/CoreSync-CRM

Java 21

Backend Engineer

- **CoreSync CRM:** Built multi-tenant B2B platform with Spring Boot 3.2 and Java 21, utilizing Spring AOP for non-intrusive domain auditing and strict logical tenant isolation.
- **NexoTrade-OS:** Implemented real-time Binance WebSocket streaming with Java 21 Virtual Threads, executing autonomous statistical arbitrage (SMA-50) with automated unit test suites.

EDUCATION, LEADERSHIP & CERTIFICATIONS

B.S. in Software Engineering — UniEVANGÉLICA (Anápolis, GO, Brazil) Feb 2025 – Dec 2028 (Expected) · 4th Semester

Vice-President, CodeUp Academic Directory · Finalist, Start Agro Innovation 2025

Certifications: Scrum Fundamentals Certified (SFC™) — SCRUMstudy (03/2026) · Technical Storytelling — UniEVANGÉLICA (2025)

Languages: English: Advanced (Scientific academic writing, technical decision explanations, verbal fluency) · Portuguese: Native